Package 'BOLDx'

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Type Package

Title Prediction and Time-to-Event Extensions of Ordered Lattice Dose Finding

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Description Research simulation tools for prediction-assisted preventive intervention, delayed toxicity outcomes, and their combination in Bayesian ordered lattice dose finding. Includes complete-follow-up BOLD and BOIN comparators, a time-to-event BOIN comparator, configurable Beta priors, sequential child-specific mixture weights, observed-history cohort dose recommendations, intervention planning flags, and Monte Carlo operating characteristics. Extends the design of Wang and Tatsuoka (2026) <doi:10.1002/sim.70456>.

License GPL (>= 3)

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BugReports https://github.com/wguo3773/BOLDx/issues

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`beta_mixture` *Inspect sequential weighted Beta components*

Description

Compute mixture weights, Beta component parameters, the mixture mean, and the overdose probability for prediction and partial-follow-up histories.

Usage

```
beta_mixture(
  a,
  b,
  q = numeric(),
  labels = rep(-1L, length(q)),
  sensitivity = 0.9,
  specificity = 0.98,
  target = 0.25
)
```

Arguments

a, b	Positive Beta parameters after resolved, nonlatent outcomes.
q	Remaining DLT-window fraction for each latent patient in [0,1]. Use 1 for an intervention patient without an observed DLT.
labels	Prediction labels: 1 positive, 0 negative, -1 uninformative.
sensitivity, specificity	Prediction operating characteristics.
target	Tail threshold for computing overdose probability.

Details

Components are weighted by sequential child-specific plug-in scores. Patient order is therefore part of the calculation. Known outcomes should first be incorporated into a and b. An observed intervention failure is a known DLT: increment a rather than branching.

Value

List containing components, mean, and overdose_probability.

Examples

```
beta_mixture(.75, 2.25, q = c(1, .5), labels = c(1L, -1L))
```

`bold_config` *Configure BOLD simulations and patient-history recommendations*

Description

The prior at dose j is Beta(prior_mean[j] * prior_pess[j], (1 - prior_mean[j]) * prior_pess[j]). True untreated risks are separate simulation inputs and are never supplied to dose-assignment calculations.

Usage

```
bold_config(
  target = 0.25,
  nmax = 21L,
  tau = 0.5,
  overdose_cutoff = 0.9,
  early_n = 12L,
  sensitivity = 0.9,
  specificity = 0.98,
  eligibility = 1,
  rho = 0.8,
  window = 3,
  accrual_rate = 2,
  arrival = c("uniform", "exponential"),
  timing = c("weibull", "uniform", "late"),
  prior_mean = NULL,
  prior_pess = 3,
  start_dose = 1L,
  pending_limit = 0.5,
  min_two = FALSE,
  gamma = overdose_cutoff
)
```

Arguments

`target`	Target untreated DLT probability, strictly between 0 and 5/7.
`nmax`	Patient cap, a positive multiple of three.
`tau`	BOLD decision-tail target in (0,1); not the DLT target.
`overdose_cutoff`	Posterior overdose elimination cutoff in (0,1).
`early_n`	Stop when this many patients have received the current dose and the next recommendation is unchanged. Use Inf to disable.
`sensitivity, specificity`	Prediction sensitivity and specificity for latent untreated DLT status, each in [0,1].
`eligibility`	Probability a patient is eligible for prediction.
`rho`	Probability an intervention prevents an otherwise-occurring DLT.
`window`	DLT assessment window, in months.

accrual_rate	Mean number of arrivals per month.
arrival	Arrival-gap distribution, uniform or exponential.
timing	Generative conditional DLT-time distribution. The working follow-up model remains uniform for all choices.
prior_mean, prior_pess	Dose-specific vectors, or scalars recycled to all doses. NULL prior_mean uses target; PESS defaults to 3.
start_dose	Starting dose index, validated against each scenario.
pending_limit	Maximum pending fraction at the current dose before suspension; defaults to 0.5.
min_two	Require two completed outcomes before escalation.
gamma	Alias for overdose_cutoff, default 0.90. Eliminate a dose when its posterior overdose probability exceeds this cutoff. If both names are supplied they must agree. Applies to simulations and recommendations.

Value

A validated configuration list of class bold_config.

Examples

```
cfg <- bold_config(nmax = 21, rho = 0.8)
cfg$target
```

---

plan_bold *Plan the next dose and prediction-guided intervention together*

Description

Combine an observed-history dose recommendation with intervention flags for incoming patients. Incoming predictions never alter the historical posterior.

Usage

```
plan_bold(
  history = NULL,
  ndoses,
  current_dose = NULL,
  method = c("bold", "prediction", "tite", "combined"),
  config = bold_config(),
  excluded = integer(),
  prediction = NULL,
  eligible = TRUE
)
```

Arguments

history, ndoses, current_dose, config, excluded	Arguments passed to recommend_dose().
method	One of bold, prediction, tite or combined.
prediction	External binary predictions for the incoming cohort, or NULL when predictions have not yet been obtained. A supplied cohort must contain three predictions; NA denotes an unavailable prediction.
eligible	Logical scalar or vector for the incoming cohort.

Details

Original BOLD and TITE-BOLD do not use preventive intervention. Prediction and combined modes flag eligible positive predictions. If predictions depend on dose, obtain them at the returned candidate dose before treating patients. This function neither trains a classifier nor supplies a final MTD declaration. Protocol-specific statistical review and validation are required before clinical application.

Value

A list with dose_recommendation, intervention_plan and ready. intervention_plan contains the candidate dose, intervene flags and status. No intervention is scheduled when dose allocation is waiting or stopped. ready is TRUE only when a dose is assigned and all required incoming prediction decisions are available. Actual interventions must subsequently be recorded in history, not inferred from these proposed flags.

Examples

```
cfg <- bold_config(gamma=.90, tau=.50)
h <- data.frame(dose=c(1,1,1), dlt=c(0,0,0), followup=c(3,3,3))
p <- plan_bold(h, 4, 1, method="prediction", config=cfg,
               prediction=c(1L,0L,0L))
p$dose_recommendation
p$intervention_plan
```

---

plan_intervention *Flag intervention eligibility from external predictions*

Description

Apply the modeled positive-prediction intervention policy. This function does not fit a classifier or perform an intervention.

Usage

```
plan_intervention(prediction, eligible = TRUE)
```

Arguments

prediction	Vector of binary predictions (0/1); NA means unavailable.
eligible	Logical vector, or scalar recycled to all predictions.

Value

A data frame containing prediction, eligible, intervene (logical, NA if an eligible patient's prediction is unavailable), and status.

Examples

```
plan_intervention(c(1L,0L,NA_integer_),eligible=c(TRUE,TRUE,FALSE))
```

---

recommend_dose *Evaluate the next cohort dose from entered patient histories*

Description

Apply the same dose-allocation rules used by the simulator to observed patient-level data. No true dose-toxicity probabilities are required.

Usage

```
recommend_dose(
  history = NULL,
  ndoses,
  current_dose = NULL,
  method = c("bold", "prediction", "tite", "combined", "boin", "tite_boin"),
  config = bold_config(),
  excluded = integer()
)
```

Arguments

history	Data frame in enrollment order with dose, dlt and followup columns. dlt=1 means a DLT has been observed; dlt=0 means none observed to date. NA is permitted only before follow-up reaches the assessment window. followup is elapsed follow-up in months. Optional prediction is 0, 1 or NA; eligible and intervened are logical columns. If omitted, eligible is inferred from prediction availability and intervened is FALSE. Use NULL for no enrolled patients.
ndoses	Number of ordered dose levels, from 2 to 10.
current_dose	Dose of the most recently enrolled cohort; required for nonempty history. For empty history, defaults to the configured start dose.
method	Design to evaluate: bold, prediction, tite, combined, boin, or tite_boin. Default is Original BOLD.
config	Design settings from bold_config().
excluded	Previously eliminated dose indices. Pass the previous result's excluded_doses to preserve irreversible elimination.

Details

Evaluation is supported at cohort boundaries of three patients. Original BOLD and prediction-assisted complete-follow-up BOLD wait until observed follow-up resolves. TITE methods use partial follow-up. Combined mode additionally waits for intervention patients to resolve.

Without intervention, a DLT or completed non-DLT follow-up replaces any pending prediction contribution. After intervention, an observed DLT is a known failure; no observed DLT leaves untreated status latent even after full follow-up. An intervention must have a positive prediction and an eligible patient. Actual intervention decisions must be explicitly recorded.

sensitivity and specificity weight available predictions. rho, eligibility probability, timing and accrual settings generate simulation histories; they do not replace observed patient records in this function. No unknown untreated outcomes should be entered. This interface implements the package's research protocol; application in a clinical trial requires protocol-specific statistical review and validation.

Value

A bold_recommendation list with action (start, escalate, stay, deescalate, wait or stop), next_dose (NA when waiting/stopped), reason, excluded_doses, dose_summary and patient_status. It is a next-cohort allocation result, not a final MTD selection.

Examples

```
h <- data.frame(dose=c(1,1,1), dlt=c(0,0,0), followup=c(3,3,3))
recommend_dose(h, ndoses=4, current_dose=1, method="bold")

hp <- transform(h, prediction=c(1L,0L,0L), eligible=TRUE,
                intervened=c(TRUE,FALSE,FALSE))
recommend_dose(hp, ndoses=4, current_dose=1, method="prediction")
```

---

simulate_bold *Simulate BOLD extensions and matched comparators*

Description

Evaluate dose-finding accuracy, enrollment, actual toxicity, allocation above the MTD, and calendar duration for BOLD extensions and their comparators.

Usage

```
simulate_bold(
  truth,
  method = c("prediction", "tite", "combined"),
  config = bold_config(),
  nsim = 100L,
  seed = 1L,
  mtd = NULL
)
```

Arguments

truth	Nondecreasing vector of 2 to 10 untreated DLT probabilities, strictly between zero and one.
method	One or more of prediction, tite, combined, bold, boin, tite_boin.
config	A configuration from bold_config().
nsim	Number of replicate trials per method.
seed	Nonnegative integer seed. The caller's random state is restored.
mtd	True optimal dose index for evaluation. NULL chooses the dose closest to target (lower index breaks ties). Set 0 for all-too-toxic; K+1 means all doses are below target and the highest dose is optimal. This value affects summaries only, never dose decisions.

Details

Fixed cohorts of three are used. Intervention failure is an observed DLT, not an extra latent branch. In combined mode, intervention follow-up resolves before the next dose decision. Mixture weights use sequential child-specific plug-in scores; the full calculation and assumptions are documented in the package's online Methods document.

Value

A bold_simulation list containing summary, trials, config, truth, mtd, and seed. Summary contains scenario-level point estimates. Optional Monte Carlo precision diagnostics are retained separately in mcse.

Examples

```
z <- simulate_bold(c(.10, .25, .45, .60),
  method = c("prediction", "bold", "boin"), nsim = 10, seed = 1)
z$summary
```

---

simulate_scenarios *Run a panel of dose-toxicity scenarios*

Description

Results include scenario-level estimates and mean/SD summaries across random scenarios within each method and MTD category. Use explicit MTD labels for all-too-toxic and below-range cases.

Usage

```
simulate_scenarios(
  scenarios,
  mtd = NULL,
  method = c("prediction", "tite", "combined"),
  config = bold_config(),
  nsim = 100,
  seed = 1
)
```

Arguments

scenarios	Numeric matrix with one nondecreasing risk vector per row.
mtd	Optional integer vector giving each scenario's optimal dose.
method, config, nsim, seed	As in simulate_bold(). Seeds increment by scenario index; methods within a scenario share the same seed.

Value

List with summary and trials data frames, each containing scenario, and aggregate containing means and between-scenario standard deviations.

Examples

```
x <- rbind(c(.1,.25,.45,.6),c(.05,.1,.25,.45))
simulate_scenarios(x, mtd=c(2,3), method="prediction", nsim=5)$summary
```

---

summarise_scenarios *Summarise operating characteristics across random scenarios*

Description

Compute the equally weighted mean and sample standard deviation of scenario-level estimates, each obtained from its simulated trial replicates.

Usage

```
summarise_scenarios(x, group_by = c("method", "mtd"))
```

Arguments

x	A data frame of scenario-level summaries, or the result returned by simulate_scenarios().
group_by	Columns defining the reporting groups; must include method. Additional columns, such as delta, can be attached to x before summarising.

Details

The SD measures between-scenario dispersion, not a standard error. For 50 scenarios with 100 replicates each, the SD is calculated from the 50 scenario-level estimates, not from the 5,000 individual trials. It includes both scenario heterogeneity and finite-replicate simulation noise. Group by MTD category, upper toxicity gap, and design settings where appropriate. A group with only one scenario has undefined SD, reported as NA. Above-MTD percentages are first patient-weighted within each scenario, then equally weighted across scenarios. Missing duration for complete-follow-up methods is retained as NA.

Value

A data frame of point-estimate means and matching *_sd columns, with n_scenarios and the minimum/maximum replicates per scenario.

Calculation

For J scenario-level estimates Q_j, the reported values are

$$\overline{Q} = J^{-1} \sum_{j=1}^{J} Q_j, \qquad SD(Q) = \sqrt{\frac{\sum_{j=1}^{J} (Q_j - \overline{Q})^2}{J-1}}.$$

Examples

```
panel <- simulate_scenarios(rbind(c(.1,.25,.45), c(.15,.25,.4)),
  mtd=c(2,2), method="prediction", nsim=5)
panel$aggregate
```

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